

F24-064-D-Doctor AI

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Chapter 1

Introduction

This project focuses on building an intelligent web application that integrates NLP and advanced AI technologies. It is primarily a **development** project with the goal of innovating within the e-mental health domain. This document is intended for various stakeholders: developers to build the system, project managers to track progress, marketing staff to promote its features, testers to ensure functionality, and documentation writers to create user guides.

1.1 Problem Statement

Among other mental health disorders anxiety, depression and stress largely impact millions of people each year. A significant percentage of people with these disorders do not receive treatment, despite their prevalence. Moreover, in Pakistan, the cultural stigma and fear of being judged makes it difficult for people to seek professional mental health assistance, which makes the treatment gap particularly acute. Our system, Doctor-AI, is designed to bridge this gap by providing a free, accessible, and stigma-free platform for mental health support. Through this platform, we aim to reduce the barriers to mental health care, making it easier for individuals to seek help without fear of judgment or financial burden.

1.2 Scope

The scope of this project focuses on the preliminary detection of anxiety, depression, and stress. The platform will enable users to register and log in with their credentials, initiate a conversation with the chatbot, and seek mental health assistance. To further this process, users will be asked to provide their medical and family history, followed by an initial assessment. Additionally, the platform will offer contact details for psychologists located in Islamabad, ensuring users have access to localized and relevant professional support. Users will also have access to a detailed report of their conversation, generated through

sentiment analysis. The chatbot's role will be limited to providing therapy or counseling and will not extend to diagnosing other mental health conditions or medical issues.

1.3 Modules

The proposed solution is an AI integrated web platform that utilizes NLP and AI to diagnose anxiety, depression, and stress. The key modules and their features include:

1.3.1 Module 1 : Web Application

This module provides the user interface for interacting with the system. It includes features such as user authentication (Login/Signup), emotional trends visualization, and 24/7 access to the trained chatbot for support.

1. User authentication (Login/Signup).
2. Visualization of emotional trends.
3. Intuitive, user-friendly interface.
4. 24/7 availability of chatbot support.

1.3.2 Module 2 : NLP Chatbot

The NLP Chatbot engages users in conversations, assesses mental health, and offers support. It records medical history and provides recommendations for self-help resources based on user input.

1. Engages users in medically crafted conversations.
2. Provides initial mental health assessments.
3. Recording medical and family history.
4. Answers questions and offers self-help resources.

1.3.3 Module 3 : Sentiment Analysis

This module analyzes user emotions in real-time during conversations. It adapts the system's behavior based on the emotional state of the user, displaying motivational texts accordingly.

1. Analyzes users' emotional states in real-time.
2. Adapts system behavior and displays inspirational texts according to emotional needs.

1.3.4 Module 4 : Resource Recommendation Engine

The recommendation engine offers tailored mental health resources (e.g., articles, exercises) based on the user's condition.

1. Recommends personalized mental health resources (articles, exercises).
2. Uses user's conversational analysis results to tailor suggestions.
3. Provides relevant materials based on specific needs.

1.3.5 Module 5 : Professionals Directory

The professionals directory provides access to a list of mental health specialists in Islamabad.

1. Maintains a directory of psychologists from Islamabad.
2. Provides contact information and consultation options.
3. Facilitates connection with healthcare providers.

1.3.6 Module 6 : Report generation and Visualisation

This module generate reports summarizing user conversations, and visualizes it in a user-friendly format.

1. Generate reports based on user conversations.
2. Reports will represent results of sentiment analysis.
3. Visualizes reports in an easy-to-understand format.
4. Aids users and professionals in understanding mental health conditions.

1.4 User Classes and Characteristics

User class	Description
General Users	General users are individuals, aged 18 and above, who seek mental health support, particularly those experiencing anxiety, depression, or stress. They interact with the AI-powered chatbot for mental health assessments, receive personalized recommendations, and may use the professional directory to find psychologists. General users prioritize anonymity, ease of use, and quick access to mental health resources.
System Administrators	System administrators are responsible for managing the technical operations of Doctor AI, including ensuring user data security, performing system updates, and maintaining the professional directory. They ensure that the system runs smoothly by overseeing both the backend (e.g., database management) and the frontend (e.g., user interface, functionality), ensuring user data is handled securely.
Project Managers and Stakeholders	Project managers and stakeholders oversee the development and progress of Doctor AI, ensuring the platform aligns with the project's objectives. They track key performance indicators such as user adoption rates, system performance, and overall platform impact. Project managers require access to system metrics, user reports, and progress dashboards to ensure the project stays on track.

Chapter 2

Project Requirements

2.1 Use-case

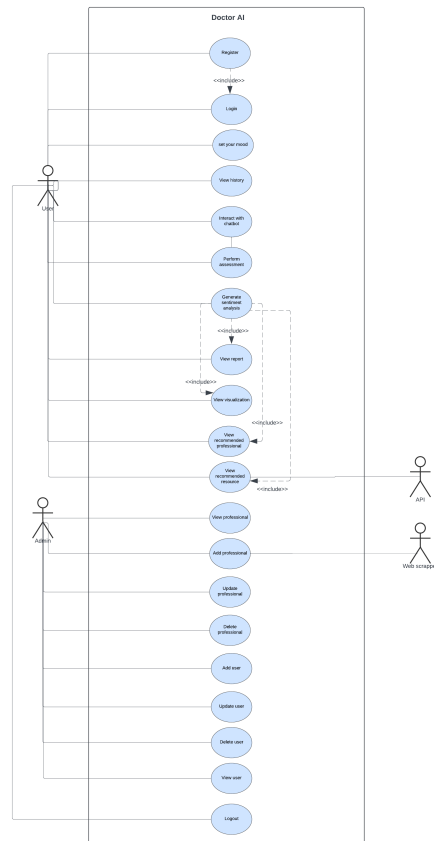


Figure 2.1: Use case diagram

2.1.1 Detailed Use Cases

1. Register

- **Actor:** User
- **Purpose:** To allow the user to create an account on the system.
- **Overview:** The user provides details such as name, email, and password to register.
- **Precondition:** The user is not logged in and does not already have an account.
- **Postcondition:** The user is registered, and account details are saved in the system.
- **Main Flow:**
 - (a) User provides registration details.
 - (b) System validates the details.
 - (c) Account is created.
 - (d) Confirmation message is sent to the user.
- **Alternative Flow:**
 - (a) If the email already exists, the system notifies the user and does not create an account.

2. Login

- **Actor:** User
- **Purpose:** To allow the user to access their account.
- **Overview:** User provides login credentials to access the system.
- **Precondition:** The user has registered.
- **Postcondition:** The user is authenticated and granted access to the system.
- **Main Flow:**
 - (a) User enters email and password.
 - (b) System verifies credentials.
 - (c) User is logged in and redirected to the dashboard.
- **Alternative Flow:**
 - (a) If credentials are invalid, the system displays an error message.

3. Set Your Mood

- **Actor:** User
- **Purpose:** To allow users to input their current emotional state and receive an inspirational message based on their mood.
- **Overview:** Users can select a mood, and the system will display an inspirational message tailored to that mood.

- **Precondition:** User must be logged in.
- **Postcondition:** If the user selects a mood, an inspirational message is displayed.
- **Main Flow:**
 - (a) User is presented with the option to set their mood.
 - (b) User selects their mood.
 - (c) System displays an inspirational message based on the selected mood.
- **Alternative Flow:**
 - (a) If the user decides not to set their mood, the system proceeds without displaying a message.

4. Interact with Chatbot

- **Actor:** User
- **Purpose:** To engage with the chatbot for conversation and for counselling.
- **Overview:** The user starts a conversation with the chatbot, which asks questions and provides assistance.
- **Precondition:** User must be logged in, registered, and must have filled out the assessment form.
- **Postcondition:** Sentiment analysis is performed only if requested
- **Main Flow:**
 - (a) User initiates the conversation.
 - (b) Chatbot responds and engages in conversation.
- **Alternative Flow:**
 - (a) If the assessment form is not completed, the system prompts the user to fill it before initiating the conversation.

5. Perform Assessment

- **Actor:** User
- **Purpose:** To collect user data through an assessment form for generating reports.
- **Overview:** The user fills out an assessment form, which is then used to generate reports based on their responses.
- **Precondition:** User must be logged in.
- **Postcondition:** After filling the form, the user can engage in a conversation with the chatbot.

- **Main Flow:**

- (a) User accesses the assessment form after logging in.
- (b) User fills out the assessment form.
- (c) System generates reports based on the user's responses.

- **Alternative Flow:**

- (a) If the user does not complete the form, the system prompts them to finish it before proceeding.

6. View History

- **Actor:** User

- **Purpose:** To allow users to review reports.

- **Overview:** Users can access a history of reports.

- **Precondition:** User must be logged in and have prior conversations.

- **Postcondition:** Report history is displayed.

- **Main Flow:**

- (a) User navigates to the history section.
- (b) The system retrieves and displays past reports.

- **Alternative Flow:**

- (a) If no history is available, the system displays a message.

7. Generate Sentiment Analysis

- **Actor:** User

- **Purpose:** To analyze user conversations and assess their emotional sentiment.

- **Overview:** The system processes user conversations to generate a sentiment analysis report.

- **Precondition:** User must have engaged in conversations with the chatbot.

- **Postcondition:** Sentiment analysis report is generated and displayed.

- **Main Flow:**

- (a) User requests to generate sentiment analysis.
- (b) System analyzes the user conversations.
- (c) Sentiment analysis report is generated.
- (d) System displays the report to the user.

- **Alternative Flow:**

- (a) User logs out.

8. View Recommended Professional

- **Actor:** User
- **Purpose:** To allow the user to view a list of available psychologists.
- **Overview:** The system shows a list of professionals for consultation.
- **Precondition:** User must be logged in.
- **Postcondition:** A list of professionals is displayed.
- **Main Flow:**
 - (a) User selects the view professional option.
 - (b) System fetches and displays professionals.
- **Alternative Flow:**
 - (a) If no professionals are available, the system shows a message of no professionals available.

9. View Report

- **Actor:** User
- **Purpose:** To allow users to view reports based on sentiment analysis and conversations.
- **Overview:** Users can access reports generated from their interactions with the system.
- **Precondition:** User must be logged in, and user must have interacted with the chatbot.
- **Postcondition:** The report is displayed.
- **Main Flow:**
 - (a) User selects the view report option.
 - (b) System displays the report.
- **Alternative Flow:**
 - (a) If no report is available, the system shows an appropriate message.

10. View Visualization

- **Actor:** User
- **Purpose:** To allow users to view data visualizations based on their emotional trends.
- **Overview:** The system generates graphs or charts representing user emotional trends over time.

- **Precondition:** User must be logged in and have data available.
- **Postcondition:** Data visualization is displayed.
- **Main Flow:**
 - (a) User selects the view visualization option.
 - (b) System generates visualizations.
 - (c) Visualizations are displayed to the user.
- **Alternative Flow:**
 - (a) If no data is available, the system shows an appropriate message.

11. View Professional

- **Actor:** Admin
- **Purpose:** To view the list of psychologists.
- **Overview:** Admin can view the list of professionals in the directory.
- **Precondition:** Admin must have access to the database.
- **Postcondition:** Professional is viewed.
- **Main Flow:**
 - (a) Admin views the list of professionals in the database.
- **Alternative Flow:**
 - (a) If the action fails, the system notifies the admin.

12. Add Professional

- **Actor:** Admin and webscrapper
- **Purpose:** To add professionals in the directory
- **Overview:** Admin can add professionals in the directory
- **Precondition:** Admin must have access to the database
- **Postcondition:** Professional is added in the list
- **Main Flow:**
 - (a) Admin adds the professional in the list
 - (b) System processes the request
 - (c) System updates the directory
- **Alternative Flow:**
 - (a) If the action fails, the system notifies the admin.

13. Update Professional

- **Actor:** Admin
- **Purpose:** To update the list of psychologists.
- **Overview:** Admin can update professionals in the directory.
- **Precondition:** Admin must have access to the database.
- **Postcondition:** Professional is updated in the directory.
- **Main Flow:**
 - (a) Admin updates the list.
 - (b) System processes the request.
 - (c) System updates the directory.
- **Alternative Flow:**
 - (a) If the action fails, the system notifies the admin.

14. Delete Professional

- **Actor:** Admin
- **Purpose:** To remove a professional from the list.
- **Overview:** Admin can remove professionals in the directory.
- **Precondition:** Admin must have access to the database.
- **Postcondition:** Professional is removed from the list.
- **Main Flow:**
 - (a) Admin can delete any record.
 - (b) System processes the request.
 - (c) System updates the directory.
- **Alternative Flow:**
 - (a) If the action fails, the system notifies the admin.

15. Add User

- **Actor:** Admin
- **Purpose:** To add a user account to the system.
- **Overview:** Admin can add user accounts in the system.
- **Precondition:** Admin must have access to the database.
- **Postcondition:** User account is added.
- **Main Flow:**
 - (a) Admin adds the user to the system.

(b) System processes the request.

(c) System confirms the action.

- **Alternative Flow:**

(a) If the action fails, the system shows an error message.

16. Update User

- **Actor:** Admin

- **Purpose:** To update user accounts.

- **Overview:** Admin can update user accounts in the system.

- **Precondition:** Admin must have access to the database.

- **Postcondition:** User account is updated.

- **Main Flow:**

(a) Admin updates the user account.

(b) System processes the request.

(c) System confirms the action.

- **Alternative Flow:**

(a) If the action fails, the system shows an error message.

17. View User

- **Actor:** Admin

- **Purpose:** To view user accounts.

- **Overview:** Admin can view user accounts in the system.

- **Precondition:** Admin must have access to the database.

- **Postcondition:** User account is viewed.

- **Main Flow:**

(a) Admin views the user account.

- **Alternative Flow:**

(a) If the action fails, the system shows an error message.

18. Delete User

- **Actor:** Admin

- **Purpose:** To remove user accounts.

- **Overview:** Admin can remove user accounts in the system.

- **Precondition:** Admin must have access to the database.
- **Postcondition:** User account is removed.
- **Main Flow:**
 - (a) Admin removes the user account.
 - (b) System processes the request.
 - (c) System confirms the action.
- **Alternative Flow:**
 - (a) If the action fails, the system shows an error message.

19. View Recommended Resources

- **Actor:** User
- **Purpose:** To recommend personalized resources based on the analysis of the user's conversation.
- **Overview:** After the conversation, the user is shown relevant resources based on the results of the analysis.
- **Precondition:** User has clicked generate sentiment analysis button.
- **Postcondition:** Links of resources are displayed.
- **Main Flow:**
 - (a) User selects the view resource option.
 - (b) System displays personalized resources.
- **Alternative Flow:**
 - (a) If no resources are available, the system shows a message.

20. Logout

- **Actor:** User
- **Purpose:** To end the session.
- **Overview:** The user or admin logs out of the system.
- **Precondition:** User or admin must be logged in.
- **Postcondition:** Session ends, and user is logged out.
- **Main Flow:**
 - (a) User selects the logout option.
 - (b) System logs the user out.
 - (c) User is redirected to the login screen.
- **Alternative Flow:**
 - (a) User goes to main dashboard page.

2.2 Functional Requirements

Following are the Functional Requirements of Doctor AI:

2.2.1 Module 1: Web Application

1. FR001: The user shall be able to register by providing personal information such as name, email and password etc.
2. FR002: Upon successful registration, the system shall display a message: “You’ve been registered.”
3. FR003: The user shall be able to log in with their registered credentials.
4. FR004: Upon successful login, the system shall display a message: “Login successful.”
5. FR005: The system shall provide an authentication mechanism for users to log in securely.
6. FR006: The system shall redirect the user to the main dashboard after successful login.
7. FR007: The user shall be presented with three options: “Set Your Mood,” “Interact with Chatbot,” and “View History.”
8. FR008: The system shall display appropriate content based on the option selected by the user.
9. FR009: The system shall allow the user to select their current emotion from a pre-defined list of emotions.
10. FR010: Based on the user’s selected emotion, the system shall retrieve an inspirational message from the database.
11. FR011: The inspirational message shall be displayed to the user after selecting their mood.
12. FR012: If the user has interacted with the chatbot before, the system shall display past conversation’s sentiment analysis reports.
13. FR013: If the user is new and has no history, the system shall display a message: “No history available.”
14. FR014: The user shall be able to log out from the system at any time by clicking the “Logout” button.

15. FR015: Upon logging out, the system shall end the session and redirect the user to the login page.

2.2.2 Module 2: Interact with Chatbot

1. FR016: Upon clicking “Interact with Chatbot,” the system shall display a form that the user must fill out before starting the conversation.
2. FR017: The form shall collect user details and any necessary medical or mental health history.
3. FR018: The user shall sign a liability form before being able to converse with the chatbot.
4. FR019: Once the form is submitted, the system shall allow the user to start a conversation with the chatbot.
5. FR020: The chatbot shall provide mental health support by responding to user queries related to mental health issues.
6. FR021: The chatbot shall engage users in a supportive conversation, offering guidance and suggestions based on their inputs.
7. FR022: The chatbot shall provide a friendly and empathetic interaction, helping users feel less lonely or confused during the conversation.
8. FR023: The user shall have the option to either log out or click “Generate Sentiment Analysis” after the conversation.

2.2.3 Module 3: Sentiment Analysis

1. FR024: Upon clicking “Generate Sentiment Analysis,” the system shall analyze the conversation data and form inputs to generate a sentiment report.
2. FR025: The sentiment analysis shall evaluate emotions such as depression, anxiety, stress or normal based on the conversation.
3. FR026: The system shall display an option for the user to “View Report” or “View Visualization” of the sentiment analysis results.

2.2.4 Module 4: View Report and Visualization

1. FR027: The user shall be able to view visualizations of their emotional trends based on the sentiment analysis.

2. FR028: The user shall be able to view a detailed report summarizing the sentiment analysis results.
3. FR029: The system shall save user reports in the database along with the user's profile.
4. FR030: The system shall ensure that only the user can view their reports, preventing access by any other users.
5. FR031: The system shall implement access control measures to guarantee that user reports remain confidential and secure.
6. FR032: Users shall be able to view their saved reports at any time through their profile.

2.2.5 Module 5: Recommendation Engine

1. FR033: After sentiment analysis, users shall receive personalized recommendations for mental health resources.
2. FR034: The user shall have the option to select "View Recommended Resources" or "View Recommended Professionals."
3. FR035: If the user selects "View Recommended Resources," the system shall search for mental health resources (e.g., articles, exercises) in real-time and provide links to the user.
4. FR036: If the user selects "View Recommended Professionals," the system shall fetch a list of recommended mental health professionals from the database.
5. FR037: The recommended professionals shall be displayed with their details.
6. FR038: The system shall store professionals' data in the database through web scraping of relevant websites.

2.2.6 Module 6: Admin Management

1. FR039: The admin shall be able to add new professionals to the database.
2. FR040: The admin shall be able to update the information of existing professionals in the database.
3. FR041: The admin shall be able to delete professionals from the database.
4. FR042: The admin shall be able to view the list of all professionals stored in the database.

5. FR043: The admin shall be able to add new users to the system with relevant details like name, email etc.
6. FR044: The admin shall be able to update user information.
7. FR045: The admin shall be able to delete users from the system if needed.
8. FR046: The admin shall be able to view a list of all users registered in the system.
9. FR047: The admin shall be able to add new inspirational messages to the database.
10. FR048: The admin shall be able to update existing inspirational messages in the database.
11. FR049: The admin shall be able to delete inspirational messages from the database.
12. FR050: The admin shall be able to view all inspirational messages stored in the database.

2.3 Non-Functional Requirements

Following are the Non-Functional Requirements of Doctor AI:

2.3.1 Security

Access to sensitive data (e.g., user reports, profiles) shall be restricted to the owner of the data and admins only.

2.3.2 Usability

The system shall provide a simple and intuitive interface, ensuring that users can interact with the platform without requiring technical knowledge. Users should be able to navigate key features such as interacting with the chatbot, viewing reports and recommendations etc.

2.3.3 Maintainability

The system code shall be well-documented to facilitate easy updates and improvements without affecting the overall functionality.

2.4 Domain Model

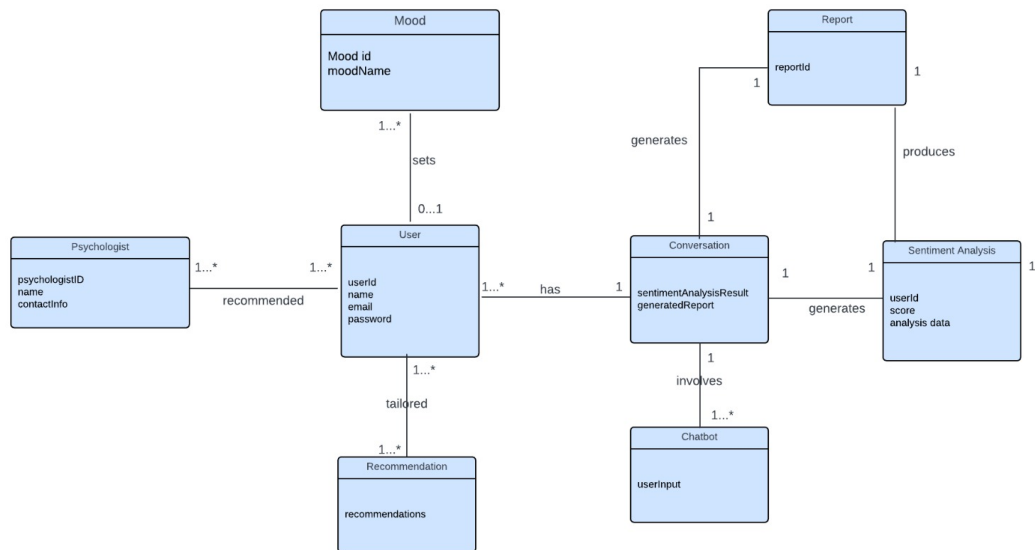


Figure 2.2: Domain Model

Chapter 3

System Overview

Doctor AI is a web-based mental health support platform that utilizes AI and Natural Language Processing (NLP) to offer users personalized mental health assessments and guidance. The platform's functionality revolves around three core features: an AI-powered chatbot, sentiment analysis, and personalized recommendations. Users can interact with the chatbot to express their emotions, ask questions, and receive mental health support. Through NLP, the chatbot evaluates the user's emotional state in real time and generates sentiment analysis, which provides insights into their mental well-being. Based on this analysis, the platform offers tailored recommendations, such as mental health exercises, relevant articles, or connections to professional mental health providers, ensuring personalized and actionable support.

3.1 Architectural Design

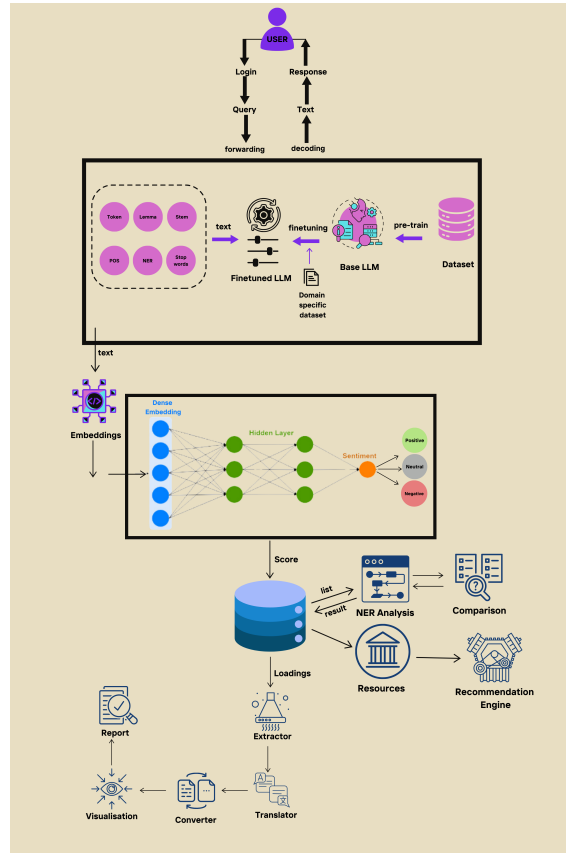


Figure 3.1: Architecture design diagram

3.1.1 User Interaction Module

- **Role:** Users interact with the system by logging in, sending queries, and receiving responses.
- **Relation:** The module forwards user queries to the text-processing module and displays the system's responses back to the user.

3.1.2 Text Processing and Fine-tuning (LLM)

- **Role:** Pre-processes the user input using tokenization, lemmatization, and fine-tuning of the language model (LLM) with a mental health dataset.
- **Relation:** Passes the processed text to the sentiment analysis module for further processing.

3.1.3 Sentiment Analysis Module

- **Role:** Converts the processed text into embeddings and analyzes the user's sentiment.
- **Relation:** Sends sentiment results to the recommendation engine to guide the next actions.

3.1.4 NER and Recommendation Engine

- **Role:** Extracts important entities (like medical terms) and compares them with a database of mental health resources to provide personalized recommendations.
- **Relation:** Uses sentiment analysis and entity recognition to offer tailored guidance and exercises.

3.1.5 Data Storage and Reporting Module

- **Role:** Stores user data and reports.
- **Relation:** Ensures user reports and data are securely saved and accessible for future retrieval of reports.

3.2 Design Models

3.2.1 Activity Diagram

The activity diagram for Doctor AI starts with the user logging in. After a successful login, the user can choose to set their mood, interact with the chatbot, or view their history. Once they interact with the chatbot, they can perform an assessment and engage in further conversation. At a decision point, the user is asked whether they want to view an analysis of their conversation. If they agree, the system generates sentiment analysis.

Following this, the user can view recommendations (either professionals or resources), their report, or a visualization of emotional trends. The diagram also shows that, after completing these tasks, the user can log out, ending their session.

This flow effectively illustrates the primary actions a user can take, moving from login through various interaction points, decision-making, and ultimately to logout.

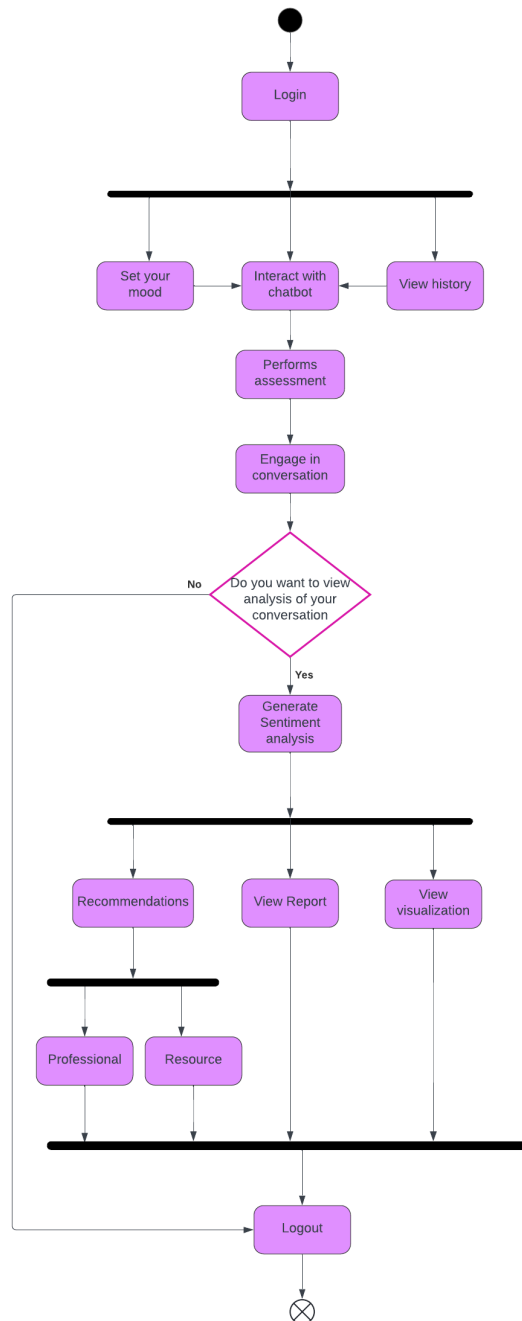


Figure 3.2: Activity Diagram

3.2.2 Data Flow Diagram

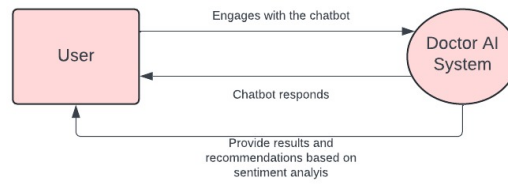


Figure 3.3: Data Design: Level 0

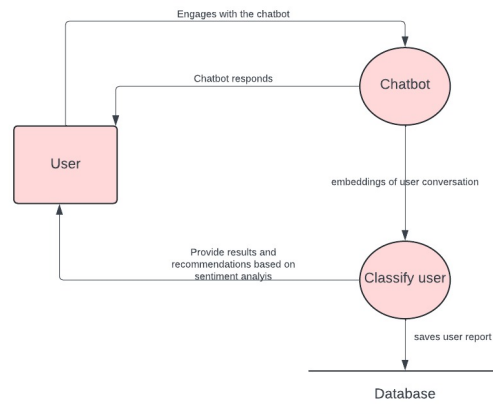


Figure 3.4: Data Design: Level 1

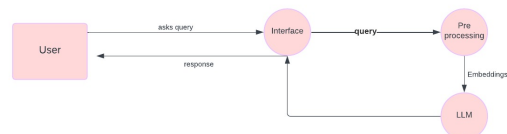


Figure 3.5: Data Design: Level 2.1



Figure 3.6: Data Design: Level 2.2



Figure 3.7: Data Design Level 2.3



Figure 3.8: Data Design Level 2.4

3.2.3 System Sequence Diagram

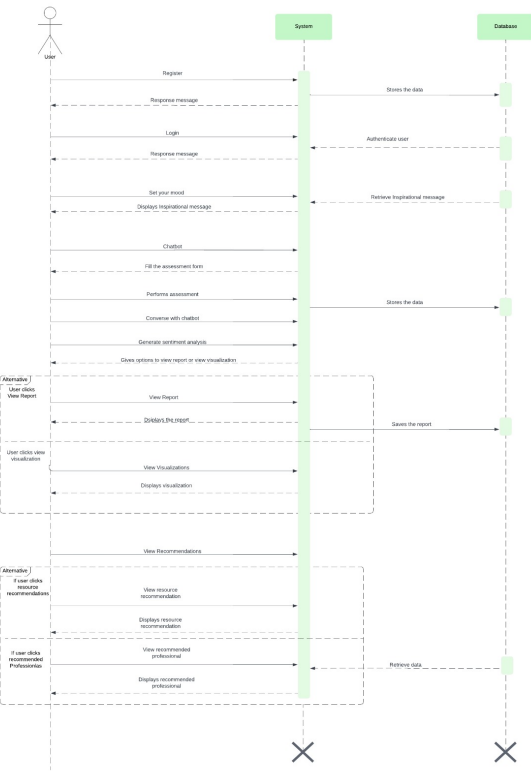


Figure 3.9: System Sequence Diagram

3.3 Data Design

The information domain of the system is effectively organized into structured data entities that facilitate smooth processing, storage, and retrieval of user-related information and professional resources. Below is an overview of how these data entities are represented and managed:

3.3.1 Data Structures and Formats

Dataset for Fine-Tuning Llama:

For fine-tuning the Llama model, we utilized a dataset in JSON format. JSON (JavaScript Object Notation) was chosen due to its lightweight nature and ease of use in representing structured data. The dataset includes two columns of context and response.

```
C:\Users\Admin\Desktop> New folder > {} cleaned_dataset.json > ...
1 [{"Context": "how are you doing today", "Response": "okay how 'bout yourself"}]
2 [{"Context": "where are you from originally", "Response": "here in california"}]
3 [{"Context": "really", "Response": "yeah"}]
4 [{"Context": "what are some things you really like about l a ", "Response": "oh well that it's big and broad there's a lot to do a lot of um"}]
5 [{"Context": "what are some things you don't really like about l a", "Response": "traffic um maybe the violence rate bad news even though you know yo"}]
6 [{"Context": "what'd you study at school", "Response": "sociology i have a um two year degree in liberal arts but my major was sociology"}]
7 [{"Context": "are you still working in that", "Response": "no i'm actually an m_t a bus operator"}]
8 [{"Context": "what's your dream job", "Response": "well since metro is steady growing um my dream job is to move up in the company and i'm about to a"}]
9 [{"Context": "that sounds like a great situation ", "Response": "thanks"}]
10 [{"Context": "how hard is that", "Response": "well it's not actually hard as right now it's easy if you take advantage of it right now 'cause they lo"}]
11 [{"Context": "do you consider yourself an introvert", "Response": "what do you mean i'm sorry"}]
12 [{"Context": "let's come back to that later", "Response": "okay"}]
13 [{"Context": "what do you do to relax", "Response": "read take a long walk hot bath meditate just close my eyes sometimes"}]
14 [{"Context": "how are you at controlling your temper ", "Response": "i'm pretty much good because see by me being a bus operator you run into circum"}]
15 [{"Context": "how hard is that", "Response": "well i look at it like it goes with i mean it comes with experience you know um you do something on"}]
16 [{"Context": "has that gotten you in trouble", "Response": "no i'm pretty good at it now"}]
17 [{"Context": "who's someone that's been a positive influence in your life", "Response": "i would say my family you know my kids you know even though"}]
18 [{"Context": "that sounds like a great situation", "Response": "thank you"}]
19 [{"Context": "do you find it easy to be a parent", "Response": "no "}]
20 [{"Context": "what's the hardest thing about being a parent", "Response": "um setting good examples you know because you're the first teacher you kno"}]
21 [{"Context": "can you give me an example of that", "Response": "well um "}]
22 [{"Context": "tell me about your kids", "Response": "i have a um four and a half year old son i have a seven year old daughter and i also have a twen"}]
23 [{"Context": "aw", "Response": "yeah i'm 'bout to be"}]
24 [{"Context": "awesome ", "Response": "yeah thank you"}]
25 [{"Context": "what's the hardest thing about being a parent", "Response": "the hardest thing is"}]
26 [{"Context": "yeah", "Response": "they can be expensive at times you know it depends on the person's situation you know the whether they have help or"}]
27 [{"Context": "what's one of your most memorable experiences ", "Response": "with my kids or just period"}]
28 [{"Context": "whatever comes to your mind", "Response": "what's memorable for me is that i'm the only one in my family with a degree so it's kinda li"}]
29 [{"Context": "wow", "Response": "and i've always kept that in the back of my mind even though everybody well pretty much my whole family have decent"}]
30 [{"Context": "that's so good to hear", "Response": "thank you"}]
31 [{"Context": "how did you feel in that moment ", "Response": "happy that i accomplished something you know just even though it may not be much but it"}]
```

Figure 3.10: Dataset Preview

3.3.2 Data Storage and Organization

- **User Registration Data:**

The system maintains a database to store user registration data, which includes user credentials. This database ensures secure access and allows users to log in and retrieve their information as needed.

- **User Reports:**

User-generated reports, including assessment results and sentiment analysis outcomes, are stored within the database. These reports can be accessed by users for self-reflection and to track their mental health progress over time. The organization of this data facilitates efficient querying and retrieval of historical assessments.

- **Inspirational Messages:**

The system includes a repository of inspirational messages tailored to users' moods and situations. These messages are linked to specific emotional states and are retrieved dynamically based on the user's selection of mood.

- **Mental Health Professionals Data:**

The system collects data about mental health professionals through web scraping

techniques. This data is organized into a structured format within the database, allowing users to easily access information about psychologists located in Islamabad. The web-scraped data includes professional qualifications, contact details etc.

Chapter 4

Conclusions

This project is set to change the mental health landscape by connecting the technologies of AI and NLP. Our goal is to deliver a unified, secure, effective, and user-centric solution that bridges the mental health treatment gap, offering accessible support in a stigma-free environment. By successfully implementing this system, we aim to empower individuals to manage their mental health with confidence and ease, making quality care more accessible in the digital age.

4.1 Professionals Onboard

1. **Dr. Rizwan Taj**

MBBS, MRCPSYCH (UK), Diploma in Management Med. Doctor (Management for Medical Doctor), Diploma in Clinical Psychiatry (Psychiatry), Diploma in Psychological Medicine (Psychological Medicine)

2. **Ayesha Wajahat**

Mindfulness Meditation Therapist (USA), Cognitive Behavioral Therapist (UK)

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